Web Configuration

Table of Contents

Chapter 1 HTTP Switch Configuration....3
1.1 HTTP Configuration....3
1.1.1 Choosing the Prompt Language....3
1.1.2 Setting the HTTP Port....3
1.1.3 Enabling the HTTP Service....3
1.1.4 Setting the HTTP Access Mode....3
1.1.5 Setting the HTTP Login Log....4
1.1.6 Setting the Maximum Number of VLAN Entries on Web Page....4
1.1.7 Setting the Maximum Number of Multicast Entries Displayed on a Web Page....4
1.2 HTTPS Configuration....4
1.2.1 It is used to set the HTTPS port....4
Chapter 2 Configuration Preparation....6
2.1 Accessing the Switch Through HTTP....6
2.1.1 Initially Accessing the Switch....6
2.1.2 Upgrading to the Web-Supported Version....6
2.2 Accessing a Switch Through Secure Links....7
2.3 Introduction of Web Interface....7
2.3.1 Top Control Bar....8
2.3.2 Navigation Bar....9
2.3.3 Configuration Area....9
2.3.4 Configuration Area....10
Chapter 3 Basic Configuration....11
3.1 IP Address Configuration....11
3.2 Hostname Configuration....12
Chapter 4 Remote Monitoring Configuration....13
4.1 SNMP Configuration....13
4.1.1 SNMP Community Management....13
4.1.2 SNMP Host Management....14
4.2 RMON....14
4.2.1 RMON Statistic Information Configuration....14
4.2.2 RMON History Information Configuration....15
4.2.3 RMON Alarm Information Configuration....16
4.2.4 RMON Event Configuration....16
Chapter 5 System Management....18
5.1 User Management....18
5.1.1 User List....18
5.1.2 Establishing a New User....19
5.1.3 User Group Management....19
5.1.4 Password Rule Management....20
5.1.5 Authentication Rule Management....21
5.1.6 Authorization Rule Management....21
5.2 Clock Management....22

5.3 Log Management..... 22
5.4 Diagnosis Tools.....23
5.5 Managing the Configuration Files.....24
5.5.1 Exporting the Configuration Information.....24
5.5.2 Importing the Configuration Information.....24
5.6 Software Management.....25
5.6.1 Backing up the IOS Software.....25
5.6.2 Upgrading the IOS Software.....25
5.7 Resuming Initial Configuration.....26
5.8 Rebooting the Device..... 26
5.9 The About Page..... 27

Chapter 1 HTTP Switch Configuration

1.1 HTTP Configuration

Switch configuration can be conducted not only through command lines and SNMP but also through Web browser. The switches support the HTTP configuration, the abnormal packet timeout configuration, and so on.

1.1.1 Choosing the Prompt Language

Up to now, switches support two languages, that is, English and Chinese, and the two languages can be switched over through the following command.

Command	Purpose
ip http language {chinese | english}	Sets the prompt language of Web configuration to **Chinese** or **English**.

1.1.2 Setting the HTTP Port

Generally, the HTTP port is port 80 by default, and users can access a switch by entering the IP address directly; however, switches also support users to change the service port and after the service port is changed you have to use the IP address and the changed port to access switches. For example, if you set the IP address and the service port to **192.168.1.3** and **1234** respectively, the HTTP access address should be changed to **http:// 192.168.1.3:1234**. You'd better not use other common protocols' ports so that access collision should not happen. Because the ports used by a lot of protocols are hard to remember, you'd better use port IDs following port 1024.

Command	Purpose
ip http port { *portNumber* }	Sets the HTTP port.

1.1.3 Enabling the HTTP Service

Switches support to control the HTTP access. Only when the HTTP service is enabled can HTTP exchange happen between switch and PC and, when the HTTP service is closed, HTTP exchange stops.

Command	Purpose
ip http server	Enables the HTTP service.
ip http timeout	Configures the timeout time of HTTP abnormal packets.

1.1.4 Setting the HTTP Access Mode

You can access a switch through two access modes: HTTP access and HTTPS access, and you can use the following command to set the access mode to **HTTP**.

Command	Purpose
ip http http-access enable	Sets the HTTP access mode.

1.1.5 Setting the HTTP Login Log

Run the following command to set the HTTP login log.

Command	Purpose
ip http log-login enable	Enables the user login log.

1.1.6 Setting the Maximum Number of VLAN Entries on Web Page

A switch supports at most 4094 VLANs and in most cases Web only displays parts of VLANs, that is, those VLANs users want to see. You can use the following command to set the maximum number of VLANs. The default maximum number of VLANs is 100.

Command	Purpose
ip http web max-vlan { *max-vlan* }	Sets the maximum number of VLAN entries displayed in a web page.

1.1.7 Setting the Maximum Number of Multicast Entries Displayed on a Web Page

A switch supports at most 100 multicast entries. You can run the following command to set the maximum number of multicast entries and Web then shows these multicast entries. The default maximum number of multicast entries is 15.

Command	Purpose
ip http web igmp-groups { *igmp-groups* }	Sets the maximum number of multicast entries displayed in a web page.

1.2 HTTPS Configuration

In order to improve the security of communications, switches support not only the HTTP protocol but also the HTTPS protocol. HTTPS is a security-purposed HTTP channel and it is added to the SSL layer under HTTP.

1.2.1 It is used to set the HTTPS port.

As the HTTP port, HTTPS has its default service port, port 443, and you also can run the following command to change its service port. It is recommended to use those ports following port 1024 so as to avoid collision with other protocols' ports.

Parameter	Remarks
ip http secure-port {*portNumber*}	Sets the HTTPS port.

Chapter 2 Configuration Preparation

2.1 Accessing the Switch Through HTTP

When accessing the switch through Web, please make sure that the applied browser complies with the following requirements:

- HTML of version 4.0
- HTTP of version 1.1
- JavaScript™ of version 1.5

What's more, please ensure that the main program file, running on a switch, supports Web access and your computer has already connected the network in which the switch is located.

2.1.1 Initially Accessing the Switch

When the switch is initially used, you can use the Web access without any extra settings:

1. Modify the IP address of the network adapter and subnet mask of your computer to **192.168.0.2** and **255.255.255.0** respectively.

2. Open the Web browser and enter **192.168.0.1** in the address bar. It is noted that **192.168.0.1** is the default management address of the switch.

3. If you use the Internet Explorer browser, you can see the dialogue similar with figure 1. Both the original username and the password are “admin”, which is capital sensitive.

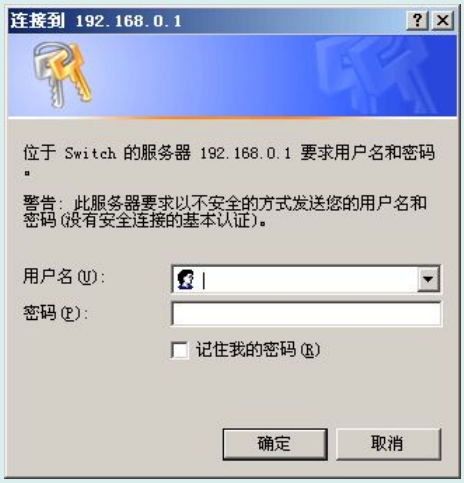


Figure 1: ID checkup of WEB login

4. After successful authentication, the systematic information about the switch will appear on the IE browser.

2.1.2 Upgrading to the Web-Supported Version

If your switch is upgraded to the Web-supported version during its operation and the switch has already stored its configuration files, the Web visit cannot be directly applied on the switch. Perform the following steps one by one to enable the Web visit on the switch:

1. Connect the console port of the switch with the accessory cable, or telnet to the management address of the switch through the computer.

2. Enter the global configuration mode of the switch through the command line, the DOS prompt of which is similar to “Switch_config#”.

3. If the management address of the switch is not configured, please create the VLAN interface and configure the IP address.

4. Enter the **ip http server** command in global configuration mode and start the Web service.

5. Run **username** to set the username and password of the switch. For how to use this command, refer to the “Security Configuration” section in the user manual.

After the above-mentioned steps are performed, you can enter the address of the switch in the Web browser to access the switch.

6. Enter **write** to store the current configuration to the configuration file.

2.2 Accessing a Switch Through Secure Links

The data between the WEB browser and the switch will not be encrypted if you access a switch through common HTTP. To encrypt these data, you can use the secure links, which are based on the secure sockets layer, to access the switch.

To do this, you should follow the following steps:

1. Connect the console port of the switch with the accessory cable, or telnet to the management address of the switch through the computer.

2. Enter the global configuration mode of the switch through the command line, the DOS prompt of which is similar to “Switch_config#”.

3. If the management address of the switch is not configured, please create the VLAN interface and configure the IP address.

4. Enter the **ip http server** command in global configuration mode and start the Web service.

5. Run **username** to set the username and password of the switch. For how to use this command, refer to the “Security Configuration” section in the user manual.

6 . Run **ip http ssl-access enable** to enable the secure link access of the switch.

7. Run **no ip http http-access enable** to forbid to access the switch through insecure links.

8. Enter **write** to store the current configuration to the configuration file.

9. Open the WEB browser on the PC that the switch connects, enter **https://192.168.0.1** on the address bar (**192.168.0.1** stands for the management IP address of the switch) and then press the **Enter** key. Then the switch can be accessed through the secure links.

2.3 Introduction of Web Interface

The Web homepage appears after login, as shown in figure 2:

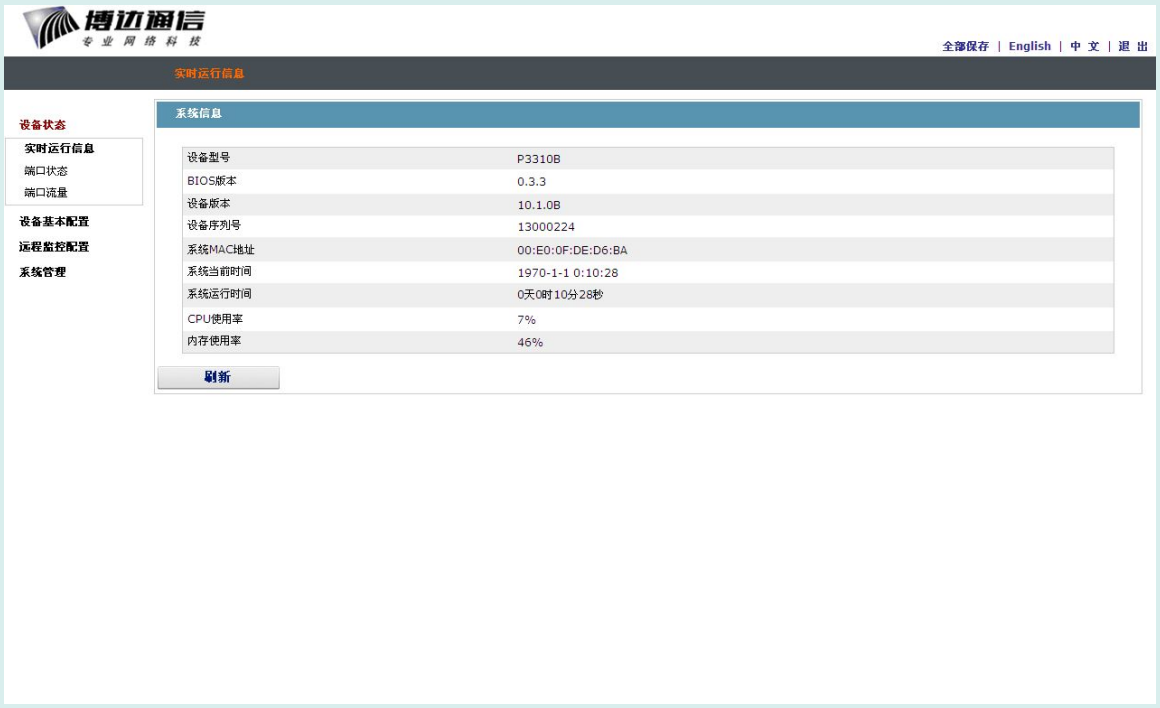


Figure 2: Web homepage

The whole homepage consists of the top control bar, the navigation bar, the configuration area and the bottom control bar.

2.3.1 Top Control Bar



Figure 3: Top control bar

Save All	Write the current settings to the configuration file of the device. It is equivalent to the execution of the **write** command. The configuration that is made through Web will not be promptly written to the configuration file after validation. If you click “Save All”, the unsaved configuration will be lost after rebooting.
English	The interface will turn into the English version.
Chinese	The interface will turn into the Chinese version.
Logout	Exit from the current login state. After you click "logout", you have to enter the username and the password again if you want to continue the Web function.

After you configure the device, the result of the previous step will appear on the left side of the top control bar. If error occurs, please check your configuration and retry it later.

2.3.2 Navigation Bar



Figure 4 Navigation bar

导航栏控制配置区中显示的内容。 The contents in the navigation bar are shown in a form of list and are classified according to types. By default, the list is located at “Runtime Info”. If a certain item need be configured, please click the group name and then the sub-item. For example, to browse the flux of the current port, you have to click “Interface State" and then “Interface Flow”.

Note:
The limited user can only browse the state of the device and cannot modify the configuration of the device. If you log on to the Web with limited user’s permissions, only “Interface State” will appear.

2.3.3 Configuration Area



Figure 5 Configuration Area

The configuration display area shows the state and configuration of the device. The contents of this area can be modified by the clicking of the items in the navigation bar.

2.3.4 Configuration Area

The configuration area is to show the content that is selected in the navigation area. The configuration area always contains one or more buttons, and their functions are listed in the following table:

Refresh	Refresh the content shown in the current configuration area.
Apply	Apply the modified configuration to the device. The application of the configuration does not mean that the configuration is saved in the configuration file. To save the configuration, you have to click “Save All” on the top control bar.
Reset	Means discarding the modification of the sheet. The content of the sheet will be reset.
New	Creates a list item. For example, you can create a VLAN item or a new user.
Delete	Deletes an item in the list.
Back	Go back to the previous-level configuration page.

Chapter 3 Basic Configuration



Figure 1 A list of basic configuration

3.1 IP Address Configuration

If you click **Basic Config -> IP Config** in the navigation bar, the **IP Configuration** page appears, as shown in figure 3.



Figure 2 IP address configuration

This IP address mainly refers to the IP address for PC communication, so you should make sure that the configured IP must be in the same network segment with PC; otherwise, PC cannot communicate with other devices.

3.2 Hostname Configuration

If you click **Basic Config -> Hostname Config** in the navigation bar, the **Hostname Configuration** page appears, as shown in figure 3.



Figure 3 Hostname configuration

The hostname will be displayed in the login dialog box.

The default name of the device is “Switch”. You can enter the new hostname in the text box shown in figure 3 and then click “Apply”.

Chapter 4 Remote Monitoring Configuration



Figure 1: Configuration tasks for remote monitoring

4.1 SNMP Configuration

If you click **Remote Monitor Config -> SNMP management** in the navigation bar, the **SNMP community and host management** page appears, as shown in figure 4.1.

4.1.1 SNMP Community Management



Figure 2 SNMP community management

On the SNMP community management page, you can know the related configuration information about SNMP community.

You can create, modify or cancel the SNMP community information, and if you click **New** or **Edit**, you can switch to the configuration page of SNMP community.



Figure 4.2 SNMP community management settings

On the SNMP community management page you can enter the SNMP community name, select the attributes of SNMP community, which include Read only and Read-Write.

4.1.2 SNMP Host Management

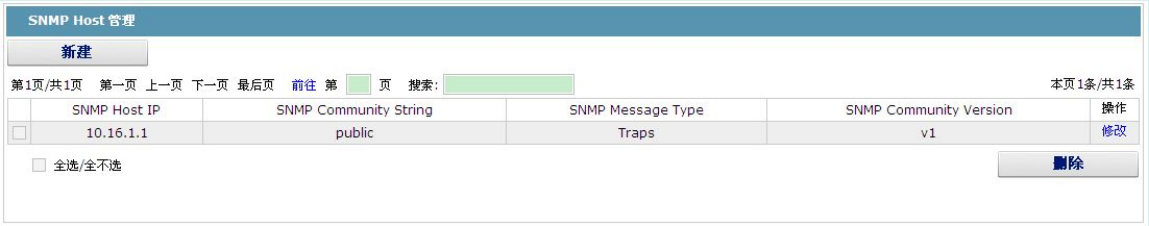


Figure 4 SNMP host management

On the SNMP community host page, you can know the related configuration information about SNMP host.

You can create, modify or cancel the SNMP host information, and if you click **New** or **Edit**, you can switch to the configuration page of SNMP host.



Figure 5 SNMP host management settings

On the SNMP host configuration page, you can enter **SNMP Host IP**, **SNMP Community**, **SNMP Message Type** and **SNMP Community Version**. **SNMP Message Type** includes **Traps** and **Informs**, and as to version 1, **SNMP Message Type** does not support **Informs**.

4.2 RMON

4.2.1 RMON Statistic Information Configuration

If you click **Remote Monitor -> RMON Statistics -> New**, the **RMON statistics** page appears.



Figure 6 Configuring the RMON statistic information

You need to set a physical port to be the reception terminal of the monitor data.

The index is used to identify a specific interface; if the index is same to that of the previous application interface, it will replace that of the previous application interface.

At present, the monitor statistic information can be obtained through the command line “show rmon statistics”, but the Web does not support this function.

4.2.2 RMON History Information Configuration

If you click **Remote Monitor -> RMON History -> New**, the **RMON History config** page appears.



Figure 7 Configuring the RMON history information

You need to set a physical port to be the reception terminal of the monitor data.

The index is used to identify a specific interface; if the index is same to that of the previous application interface, it will replace that of the previous application interface.

The sampling number means the items that need be reserved, whose default value is 50.

The sampling interval means the time between two data collection, whose default value is 1800s.

At present, the monitor statistic information can be obtained through the command line “show rmon history”, but the Web does not support this function.

4.2.3 RMON Alarm Information Configuration

If you click **Remote Monitor -> RMON Alarm -> New**, the **RMON Alarm config** page appears.



Figure 8 Configuring the RMON alarm information

The index is used to identify a specific alarm information; if the index is same to the previously applied index, it will replace the previous one.

The MIB node corresponds to OID.

If the alarm type is **absolute**, the value of the MIB object will be directly monitored; if the alarm type is **delta**, the change of the value of the MIB object in two sampling will be monitored.

When the monitored MIB object reaches or exceeds the rising threshold, the event corresponding to the index of the rising event will be triggered.

When the monitored MIB object reaches or exceeds the falling threshold, the event corresponding to the index of the falling event will be triggered.

4.2.4 RMON Event Configuration

If you click **Remote Monitor -> RMON Event -> New**, the **RMON Event config** page appears.



Figure 9 RMON event configuration

The index corresponds to the rising event index and the falling event index that have already been configured on the **RMON alarm config** page.

The owner is used to describe the descriptive information of an event.

"Enable log" means to add an item of information in the log table when the event is triggered.

“Enable trap” means a trap will be generated if the event is triggered.

Chapter 5 System Management



Figure 1 Navigation list of System Management

5.1 User Management

5.1.1 User List

If you click **System Management -> User Manage**, the **User Management** page appears.



Figure 2 User list

You can click “New” to create a new user.

To modify the permission or the login password, click “Edit” on the right of the user list.

Note:
1. Please make sure that at least one system administrator exists in the system, so that you can manage the devices through Web.
2. The limited user can only browse the status of the device.

5.1.2 Establishing a New User

If you click “New” on the **User Management** page, the **Creating User** page appears.



Figure 3 Creating new users

In the “User name” text box, enter a name, which contains letters, numbers and symbols except “?”, “\”, “&”, “#” and the "Space" symbol. \ " & #和空格以外的字符。

In the “Password” textbox enter a login password, and in the “Confirming password” textbox enter this login password again.

5.1.3 User Group Management

Click the **User group management** tab to enter the **User group management** page.



Figure 4 User group list

You can click **New** to create a new user group.

You can also click **Delete** to remove a user group.



Figure 5: User group configuration

The name of a newly created user group must never be used before. However, the password rule name, the authentication rule name and the authorization rule name must be already created, or the newly created user group cannot work normally.

5.1.4 Password Rule Management

Click the **Password rule management** tab to enter the **Password rule management** page.



Figure 6: Password rule management list

You can click **New** to create a new password rule.

You can also click **Delete** to remove a password rule.



Figure 7: Password rule configuration

You can set some password rules on this page, for example, you can set whether a password can be given a same name of a user and whether a password must contain numbers, capital-sensitive letters and special characters, and you can also set the minimum length and validity term of a password.

After a rule is created, you can apply it to user management. When the configured password complies with the password rule, the user is in a state of invalid password; when configured password complies with the password rule, the user is in a state of normal operation.

5.1.5 Authentication Rule Management

Click the **Authentication rule management** tab to enter the **Authentication rule management** page.



Figure 8: Authentication rule management list

You can click **New** to create a new authentication rule.

You can also click **Delete** to remove an authentication rule.



Figure 9: Authentication rule configuration

The maximum number of retries and the retry period should be set together or not be set together.

5.1.6 Authorization Rule Management

Click the **Authorization rule management** tab to enter the **Authorization rule management** page.

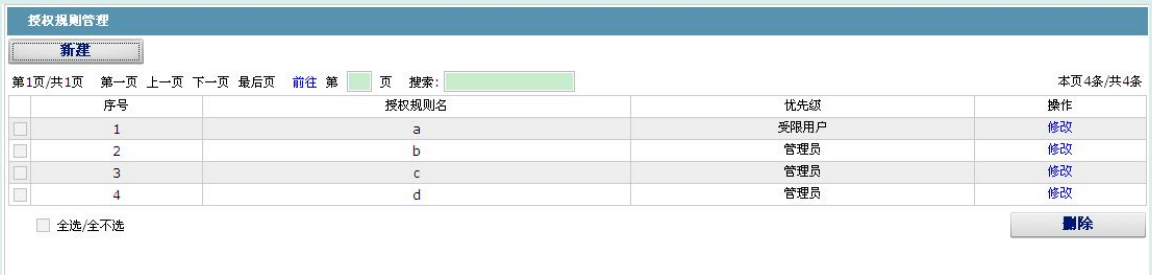


Figure 10: Authorization rule management list

You can click **New** to create a new authorization rule.

You can also click **Delete** to remove an authorization rule.



Figure 11: Authorization rule configuration

The authorization rule decides whether you are an administrator or a limited user. If you are a limited user, you cannot perform a lot of Web operations and only have the query permission.

5.2 Clock Management

If you click **System Management -> Clock Management**, the **Time Setting** page appears.



Figure 12 Clock management

To refresh the clock of the displayed device, click “Refresh”.

In the “Select Time-Zone” dropdown box select the time zone where the device is located. When you select “Set Time Manually”, you can set the time of the device manually. When you select “Network Time Synchronization”, you can designate 3 SNTP servers for the device and set the interval of time synchronization.

5.3 Log Management

If you click **System Management -> Log Manage**, the **Log Management** page appears.



Figure 13 Log management

If “Enabling the log server” is selected, the device will transmit the log information to the designated server. In this case, you need enter the address of the server in the “Address of the system log server” textbox and select the log's grade in the “Grade of the system log information” dropdown box.

If “Enabling the log buffer” is selected, the device will record the log information to the memory. By logging on to the device through the Console port or Telnet, you can run the command “show log” to browse the logs which are saved on the device. The log information which is saved in the memory will be lost after rebooting. Please enter the size of the buffer area in the “Size of the system log buffer” textbox and select the grade of the cached log in the “Grade of the cache log information” dropdown box.

5.4 Diagnosis Tools

If you click **Diagnosis Tools -> Ping**, the **Ping** page appears.



Figure 14 Ping

Ping is used to test whether the switch connects other devices.

If a Ping test need be conducted, please enter an IP address in the “Destination address” textbox, such as the IP address of your PC, and then click the “PING” button. If the switch connects

your entered address, the device can promptly return a test result to you; if not, the device will take a little more time to return the test result.

"Source IP address" is used to set the source IP address which is carried in the Ping packet.

"Size of the PING packet" is used to set the length of the Ping packet which is transmitted by the device.

5.5 Managing the Configuration Files

If you click **System Management -> Configuration file**, the **Configuration file** page appears.

5.5.1 Exporting the Configuration Information

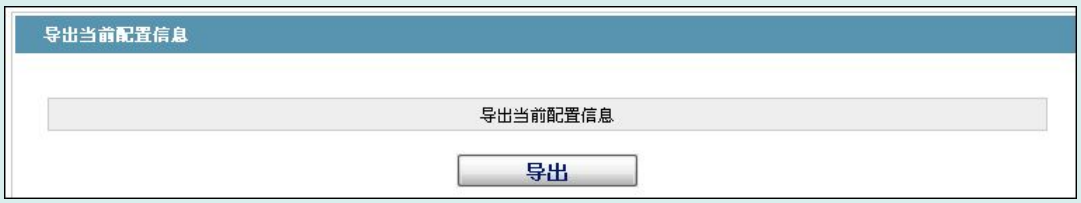


Figure 15 Exporting the configuration file

The current configuration file can be exported, saved in the disk of PC or in the mobile storage device as the backup file.

To export the configuration file, please click the "Export" button and then select the "Save" option in the pop-up download dialog box.

The default name of the configuration file is "startup-config", but you are suggested to set it to an easily memorable name.

5.5.2 Importing the Configuration Information



Figure 16 Importing the configuration files

You can import the configuration files from PC to the device and replace the configuration file that is currently being used. For example, by importing the backup configuration files, you can resume the device to its configuration of a previous moment.

Note:
1. Please make sure that the imported configuration file has the legal format for the configuration file with illegal format cannot lead to the normal startup of the device.

2. If error occurs during the process of importation, please try it later again, or click the "Save All" button to make the device re-establish the configuration file with the current configuration, avoiding the incomplete file and the abnormality of the device.
3. After the configuration file is imported, if you want to use the imported configuration file immediately, do not click "Save All", but reboot the device directly.

5.6 Software Management

If you click **System Management -> Software Upgrade**, the software management page appears.

5.6.1 Backing up the IOS Software



Figure 17 Backing up IOS

On this page the currently running software version is displayed. If you want to backup IOS, please click "Backing up IOS"; then on the browser the file download dialog box appears; click "Save" to store the IOS file to the disk of the PC, mobile storage device or other network location.

Note:
The default name of the IOS file is "switch.bin", but you are recommended to modify this name to a name that is apt to be read and found.

5.6.2 Upgrading the IOS Software

Note:
1. Please make sure that your upgraded IOS matches the device type, because the matched IOS will not lead to the normal startup of the device.
2. The upgrade of IOS probably takes one to two minutes; when the "updating" button is clicked, the IOS files will be uploaded to the device.
3. If errors occur during upgrade, please do not restart the device or cut off the power of the device, or the device cannot be started. Please try the upgrade again.
4. After the upgrade please save the configuration and then restart the device to run the new IOS.

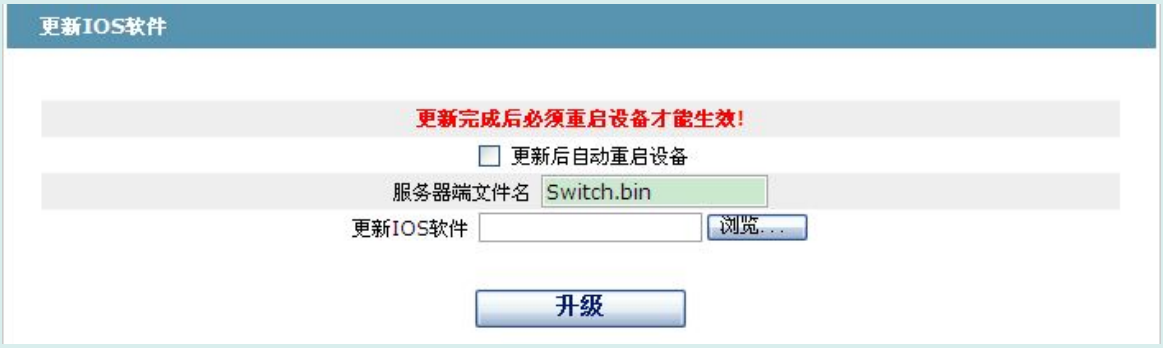


Figure 18 Upgrading the IOS software

The upgraded IOS is always used to solve the already known problems or to perfect a specific function. If you device run normally, do not upgrade your IOS software frequently.

If IOS need be upgraded, please first enter the complete path of the new IOS files in the textbox on the right of "Upgrading IOS", or click the "Browsing" button and select the new IOS files on your computer, and then click "Updating".

5.7 Resuming Initial Configuration

If you click **System Management -> Resume Config**, the **Resuming the original configuration** page appears.



Figure 19 Resuming the original configuration

Note:
1. If you click the "Resume" button, the current configuration will be replaced by the original configuration, which will take effect after rebooting.
2. Before rebooting the device still works under the current configuration, and if you click "Save All" at the moment, the current configuration will replace the original configuration. The original configuration, therefore, cannot take effect after rebooting.
3. After the rebooting is done and the original configuration takes effect, the Web access of the device will be automatically started. The address of Vlan 1 is 192.168.0.1/255.255.255.0, and the username and password are both "admin".

To resume the original configuration, click "Resume" and then reboot the device.

5.8 Rebooting the Device

If you click **System Management -> Reboot Device**, the **Rebooting** page appears.



Figure 20 Rebooting the device

If the device need be rebooted, please first make sure that the modified configuration of the device has already been saved, and then click the “Reboot” button.

5.9 The About Page

If you click **System Management -> About**, the **About** page appears.



Figure 21 “About” Page